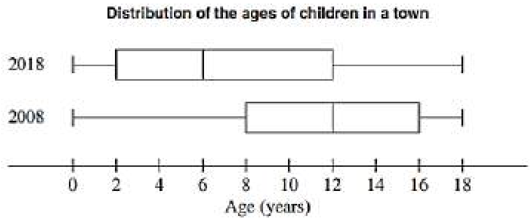
Descriptive Statistics Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 The diagram shows the distribution of the ages of children in a town in 2008 and 2018. 2



children aged 12-18 in 2008 $\frac{1}{2}$ of the total

$\therefore$ no. in 2008 $= \frac{1750}{2} = 875$

In 2018 children 12-18 is $\frac{1}{4}$ of the total

$\therefore$ children 0-18 in 2018 $= 4 \times 875 = 3500$

In 2008 there were 1750 children aged 0-18 years. The number of children aged 12-18 years was the same in both 2008 and 2018. How many children aged 0-18 were there in 2018?

2 The number of year 12 students who completed 30 minutes of exercise for each of the past ten days was as follows: 10, 11, 13, 17, 13, 15, 17, 28, 19, 15

(i) What is the mean? Answer correct to one decimal place. 1

Using the statistic mode on the calculator.

$\bar{x} = 15.8$

(ii) Find the interquartile range? 1

Using the statistic mode on the calculator.

$\text{IQR} = Q_3 - Q_1 = 17 - 13 = 4$

(iii) Is 28 an outlier for this set of data? Justify your answer with calculations. 2

$\text{Upper limit} = Q_3 + 1.5 \times \text{IQR} = 17 + 1.5 \times 4 = 23$

$\therefore$ 28 is an outlier as it is above the upper limit of 23.

3 The number of year 12 students who completed 30 minutes of exercise for each of the past ten days was as follows: 10, 11, 13, 13, 15, 15, 17, 17, 19, 28. Is 28 an outlier for this set of data? Justify your answer with calculations. 2

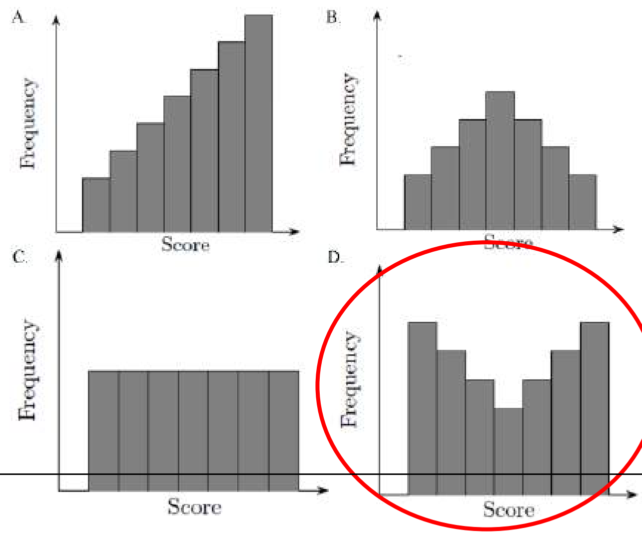
$IQR = Q_3 - Q_1 = 17 - 13 = 4$

Outlier upper limit $= Q_3 + 1.5 \times IQR$

$= 17 + 1.5 \times 4 = 23$

$\therefore$ 28 is an outlier as it is above the upper limit of 23.

4 Which of the following graphs shows data with the largest standard deviation? 1



Answer: **D** due to a lot of low scores and high scores the distance between the mean and each of the individual scores will be greater. Hence greater Standard deviation.

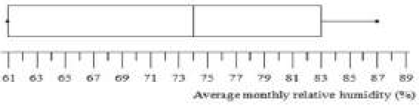
5 The average monthly relative humidity (in %) of city A is shown in the stem-and-leaf plot.

Stem	Leaf			
6	1	1	1	2
7	3	5	8	
8	3	7	7	

(i) Find the median and the inter-quartile range. 2

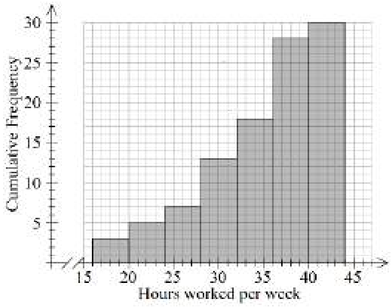
$$\text{Median} = \left(\frac{73+75}{2}\right)\% = 74\%,\quad Q_1 = 61\%, Q3 = 83\%$$

$$\therefore \text{ Inter-quartile range} = (83-61)\% = 22\%$$



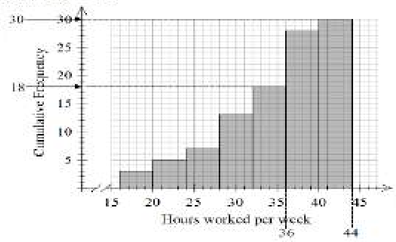
(ii) Draw a box-and-whisker plot to represent the data. 2

6 Cole is designing a survey to ask his co-workers about their job satisfaction. One of Cole's survey questions asked how many hours each respondent works at the company. The results are shown in the cumulative frequency histogram below.



(i) What is the range of responses that gave the greatest 40% of hours worked? 1

40% × 30 = 12



Top 12 respondents are from 18 to 30, which are the top two classes. These respondents worked 36 to 44 hours per week.

(ii) Use the classes in the cumulative frequency histogram to estimate the mean hours worked by the respondents surveyed. 3

Class centres are the middle of each bar of the histogram. The number of respondents in each class is the change in height for each bar of the histogram.

Sum of scores: $18 \times 3 + 22 \times 2 + 26 \times 2 + 30 \times 6 + 34 \times 5 + 38 \times 10 + 42 \times 2 = 964$

Mean: $\frac{964}{30} = 32.1$

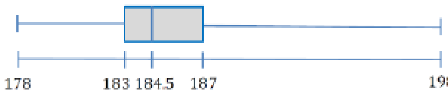
7 The heights, in cm, of a squad of 10 basketballers are listed below in ascending order:

178,180,183,183,184,185,187,187,190,198

(i) What is the median height? $\text{Median} = \frac{1}{2}(184 + 185) = 184.5$ 1

(ii) Find Q_1 and Q_3 and hence the interquartile range. $Q_1 = 183$ and $Q_3 = 187$ so $\text{IQR} = Q_3 - Q_1 = 4$. 1

(iii) Draw a box plot of the given data. 1



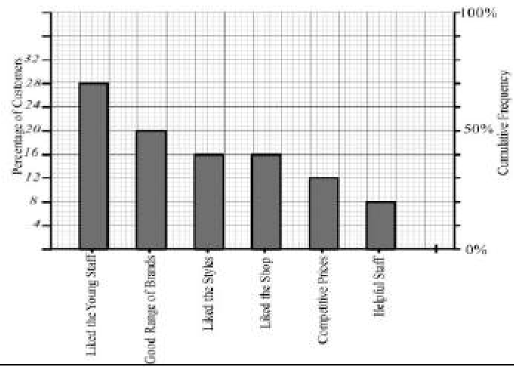
(iv) Determine whether there are any outliers amongst the heights using the criterion that an outlier is a score below $Q_1 - 1.5 \times IQR$ or above $Q_3 + 1.5 \times IQR$. 1

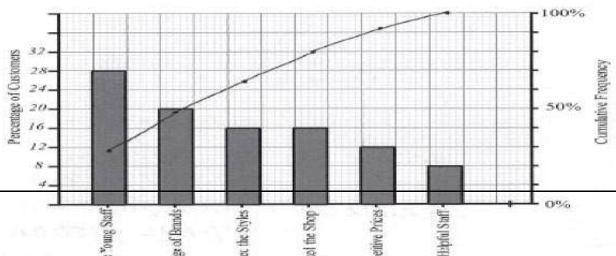
$Q_1 - 1.5 \times \text{IQR} = 183 - 1.5 \times 4 = 177$

$Q_3 + 1.5 \times \text{IQR} = 187 + 1.5 \times 4 = 193$

So 198 is an outlier.

8 The chart below shows the reasons that 25 customers gave for shopping at a local clothing store. Draw the Pareto line on the chart above. 2





9 The number of students absent from Year 12 for the past nine days was as follows:

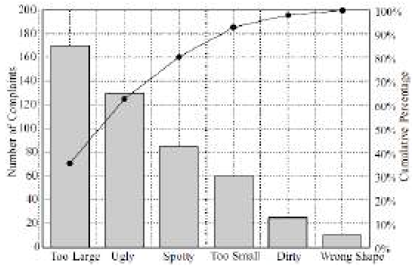
14, 17, 13, 16, 17, 12, 11, 28, 19

(i) Find the standard deviation. Answer correct to one decimal place. $\sigma_x = 4.8$ 1

(ii) Find the lower quartile and the upper quartile. Lower quartile is 12.5 Upper quartile is 18 1

(iii) Find the interquartile range. IQR = 18 − 12.5 = 5.5 1

10 The number of complaints for the return of a particular item is shown in the Pareto chart below. What percentage of the total number of complaints do the three largest complaints account for? 80% 1



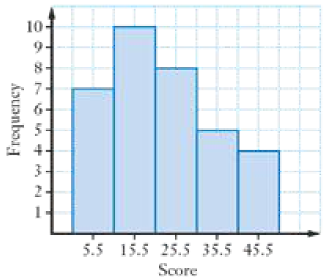
11 The mean of a set of data is 14 and the standard deviation is 2.1. If each score in the data set is increased by 4, what would happen to the mean and standard deviation? 1

The mean will increase by 4 and the standard deviation will not change

12 The volume of traffic on a stretch of highway was measured and the results are shown in the table.

Volume/min	Frequency
1–10	7
11–20	10
21–30	8
31–40	5
41–50	4

(i) Draw a histogram to show this data. 1



(ii) What was the percentage volume of traffic between 21 and 40 minutes? 1

(ii). $\frac{8+5}{7+10+8+5+4} \times 100$

$= \boxed{38.2\%}$

Descriptive Statistics Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 As winter progresses the virus spreads much more and the health authorities decide they want to stop the virus and have been given a new medication to trial. The two-way table shows the number of people in a trial.

	Taking Medication	Control Group
Virus	204	205
No Virus	212	209

(i) What percentage of people in the trial had the virus? 1

Total people = 204 + 212 + 205 + 209 = 830

$$\text{Percentage of People with Virus} = \frac{204+205}{830} \times 100 = 49\%$$

(ii) What percentage of people in the control group had the virus? 1

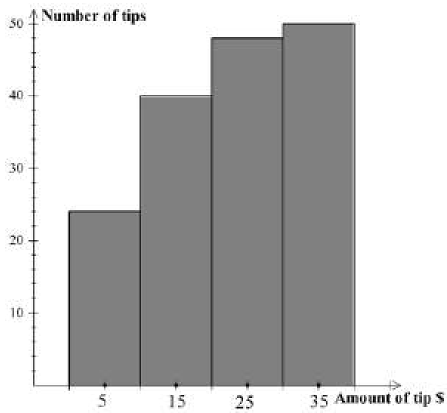
$$\text{Percentage of control group with virus} = \frac{205}{414} \times 100 = 49.5\% \approx 50\%$$

(iii) Determine if it is worth the health authorities using this new medication? 1

$$\text{Percentage of medication group with virus} = \frac{204}{416} \times 100 = 49.04\% \approx 50\%$$

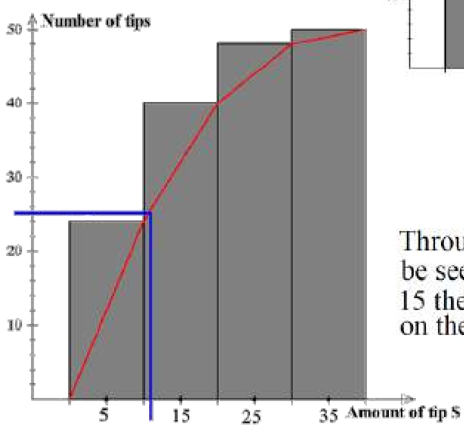
It is not worth using the medication, since almost the same percentage of the virus occurs whether on medication or not.

2 Peter works in a restaurant, and each customer can leave him a tip, some extra money, for good service. On one day, Peter earns 50 tips, as shown in the cumulative histogram:



One of the tips was incorrectly listed as $15 instead of $25. Giving reasons, explain what effect this would have on each of the following statistics:

(i) The median 1

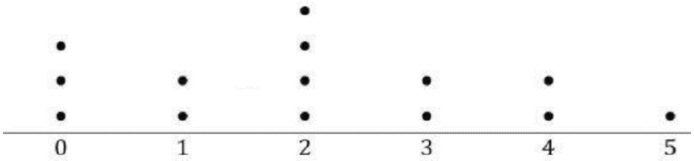


Through use of the OGIVE, it can be seen that the median is lower than 15 therefore the error has no effect on the median.

(ii) The mean 1

Since the mean calculation uses the specific value of each tip, increasing a single tip from $15 to $25 must increase the mean by $\frac{\$10}{50} = \$0.20$.

3 A sample of 14 people were asked to indicate the time (in hours) they had spent watching television on the previous night. The results are displayed in the dot plot below: 3



Find the mean and sample standard deviation of these times. Give your answers correct to one decimal place.

total point 14.

x	0	1	2	3	4	5
f	3	2	4	2	2	1
p	$\frac{3}{14}$	$\frac{2}{14}$	$\frac{4}{14}$	$\frac{2}{14}$	$\frac{2}{14}$	$\frac{1}{14}$

$$\bar{x} = E(X) = \sum xp$$
$$= 0\times\frac{3}{14} + 1\times\frac{2}{14} + 2\times\frac{4}{14} + 3\times\frac{2}{14} + 4\times\frac{2}{14} + 5\times\frac{1}{14} \quad 1$$
$$= 2\tfrac{3}{14} \simeq 2.1$$
$$\sigma = \sqrt{Var(X)} = \sqrt{E(X^2) - \mu^2} = \sqrt{\left(0 + \frac{2}{14} + 4\times\frac{4}{14} + 9\times\frac{2}{14} + 16\times\frac{2}{14} + 25\times\frac{1}{14}\right) - 2.1^2}$$
$$= 1.5 \quad 1$$

$\boxed{3}$

4 A cumulative frequency table for a data set is shown.

Score	Cumulative frequency
1	6
2	10
3	17
4	20
5	28
6	40
7	49
8	62
12	63

(i) What is the interquartile range of this data set? 2

$$\frac{1}{4}(63+1) = 16 \Rightarrow Q_1 = 3 \text{ (16th score)}$$
$$\frac{3}{4}(63+1) = 48 \Rightarrow Q_3 = 7 \text{ (48th score)}$$
$$IQR = Q_3 - Q_1 = 4$$

(ii) Explain if this data set contains an outlier. 1

$$Q_1 - 1.5\times IQR = 3 - 6 = -3$$
$$Q_3 + 1.5\times IQR = 7 + 6 = 13$$

There is no score less than -3 or more than 13.

Therefore data set does not contain any outlier.

5 The number of different dress sizes in Huang's Sportwear shop was counted in a stocktake, and the results set out in a table.

Size	Frequency
8	12
10	23
12	20
14	21
16	13
18	11

(i). How many dresses were counted? 1

Total = 12 + 23 + 20 + 21 + 13 + 11 = 100

(ii). What percentage of dresses in the store were size 12? 1

$\frac{20}{100} \times 100 = 20\%$

(iii). Find the median dress size. 1

$\frac{100}{2} = 50$ The 50^{th} size is 12

(iv). Find the 3rd quartile. 1

75% of 100 = 75.

The 75^{th} size is 14.

6 Class A gas 24 students and achieved a mean on an assessment task of 75.5%. Class B has 28 students and achieved a mean on the same assessment task of 80.5%. What was the mean mark for both classes? Answer correct to one decimal place. 2

Class A total number of marks $75.5 \times 24 = 1812.$

Class B total number of marks $80.5 \times 28 = 2254$

$$\text{Mean} = \frac{1812 + 2254}{24 + 28} = 78.1923\ldots\% \approx 78.2\%$$

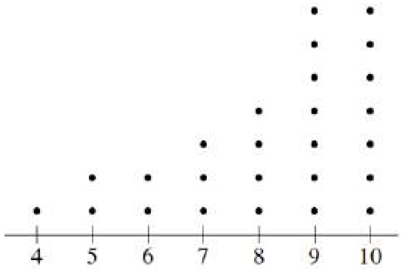
$\therefore$Mean mark for both classes is 78.2%

7 A grouped data frequency table is shown. What is the mean for this set of data? 2

Class interval	*Frequency*
1–5	3
6–10	6
11–15	8
16–20	9

$$\frac{3\times3+8\times6+13\times8+18\times9}{3+6+8+9} \approx 12.4$$

8 A set of data is displayed in this dot plot. Describe the distribution of this data set. 1



Negatively skewed

9 Consider the following data set: 1 5 9 10 15 3

Suppose a new value, x, is added to this dataset, giving the following:

1 5 9 10 15 x

It is known that x is greater than 15. It is also known that the difference between the means of the two datasets is equal to ten times the difference between the medians of the two datasets. Calculate the value of x.

Mean of first set of data $= 8$

Mean of second set of data $= \dfrac{40+x}{6}$

Median of first set of data $= 9$

Median of second set of data $= 9.5$

$$\frac{40+x}{6} - 8 = 10(9.5-9)$$

$$\frac{40+x}{6} - 8 = 5$$

$$\frac{40+x}{6} = 13$$

$$40 + x = 13 \times 6$$

$$x = 38$$

10 A experimenter records the following data:

4 6 8 8 9 10 12 12 17 24

(i) Use the standard criterion for outliers on your reference sheet to determine any outliers in the data. 3

IQR = 12−8 = 4

Outlier if > 12+1.5×4 = 18

or if < 8−1.5×4 = 2

thus 24 is an outlier

(ii) Find the mean and variance for the dataset. 2

From a calculator, $\bar{x} = 11$

$\sigma_n \doteqdot 5.5$

and variance $= \sigma_n^2 = 30.4$

(iii) Write down a dataset with the same variance but mean 15. 2

Add 4 to all values

8 10 12 12 13 14 16 16 21 28